Lesson 21: Fundamental Theorems of Functional Analysis

AI/ML Mathematics

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1 Motivation and Overview

Lesson 20 developed the rich geometry of Hilbert spaces: projections, Fourier expansions, and the self-duality $H \cong H^*$. However, many spaces that arise in reinforcement learning and optimization—spaces of bounded functions, L^p spaces with $p \neq 2$, spaces of measures—are Banach spaces that do not carry an inner product. The present lesson establishes the four fundamental theorems that govern the structure of operators and functionals on general Banach spaces:

1. The **Hahn–Banach Theorem**, which guarantees that bounded linear functionals can be extended from subspaces to the whole space without increasing the norm.
2. The **Uniform Boundedness Principle** (Banach–Steinhaus), which converts pointwise bounds on a family of operators into a uniform bound on their norms.
3. The **Open Mapping Theorem**, which states that surjective bounded linear operators between Banach spaces are open maps.
4. The **Closed Graph Theorem**, which provides a convenient criterion for verifying that a linear operator is bounded.

These theorems, together with the Baire Category Theorem (proved in Lesson 18), form the structural backbone of linear functional analysis.

Why the Fundamental Theorems Matter for RL

1. **Hahn–Banach and constrained MDPs:** In constrained Markov decision processes, the existence of Lagrange multipliers (dual variables) for constraint satisfaction is guaranteed by the Hahn–Banach theorem applied to the space of value functions. Separation arguments based on Hahn–Banach also underlie duality results in linear programming formulations of MDPs [Puterman, 2014].
2. **Uniform Boundedness and iterative algorithms:** When a sequence of operators (e.g., truncated Bellman operators, finite-sample value function approximations) converges pointwise, the Uniform Boundedness Principle guarantees that the operator norms remain uniformly bounded. This is crucial for exchanging limits in convergence proofs.
3. **Closed Graph Theorem and solution maps:** The map from problem parameters (e.g., transition kernels, reward functions) to optimal value functions is often linear on appropriate spaces. The Closed Graph Theorem provides a way to verify that such solution maps are continuous—hence stable under perturbations—without explicit norm estimates.

Prerequisites. Banach spaces, bounded linear operators, dual spaces (Lessons 15–17). The Baire Category Theorem (Lesson 18). Hilbert spaces and orthogonal projections (Lessons 19–20) for motivating examples.

Outline. Section 2 lists references. Section 3 proves the Hahn–Banach theorem and its consequences. Section 4 proves the Uniform Boundedness Principle. Section 5 proves the Open Mapping Theorem and the Bounded Inverse Theorem. Section 6 proves the Closed Graph Theorem. Section 7 provides exercises. Section 8 summarizes the key results.

2 References

The Hahn–Banach theorem is treated in Kreyszig [1989], Section 4.2; Rudin [1991], Chapter 3; and Brézis [2011], Chapter 1. The Baire Category Theorem and Uniform Boundedness Principle are covered in Kreyszig [1989], Sections 4.7–4.9, and Rudin [1991], Chapter 2. The Open Mapping and Closed Graph theorems appear in Kreyszig [1989], Sections 4.12–4.13; Brézis [2011],

Chapter 2; and Rudin [1991], Chapter 2. A historical perspective is provided by Dunford and Schwartz [1988], Chapter II. The original contraction mapping theorem appears in Banach [1922].

3 The Hahn–Banach Theorem

Definition 3.1 (Sublinear Functional)**.** A function $p\colon X \to \mathbb{R}$ on a real vector space X is *sublinear* if:

(i) $p(\alpha x) = \alpha\, p(x)$ for all $\alpha > 0$ and $x \in X$ (positive homogeneity).

(ii) $p(x + y) \le p(x) + p(y)$ for all $x, y \in X$ (subadditivity).

Theorem 3.2 (Hahn–Banach Theorem, Real Case)**.** *Let X be a real vector space, $p\colon X \to \mathbb{R}$ a sublinear functional, and $Y \subset X$ a subspace. If $f\colon Y \to \mathbb{R}$ is a linear functional with $f(y) \le p(y)$ for all $y \in Y$, then there exists a linear functional $F\colon X \to \mathbb{R}$ extending f (i.e., $F|_Y = f$) such that $F(x) \le p(x)$ for all $x \in X$.*

Proof. **Step 1: Extension by one dimension.** Let $x_0 \in X \setminus Y$ and $Y_1 = Y + \mathbb{R}x_0 = \{y + tx_0 : y \in Y,\, t \in \mathbb{R}\}$. We seek $c = F(x_0) \in \mathbb{R}$ such that $F(y + tx_0) = f(y) + tc$ satisfies $F \le p$ on Y_1.

For $t > 0$: we need $f(y) + tc \le p(y + tx_0)$, i.e., $c \le \frac{1}{t}[p(y + tx_0) - f(y)] = p(y/t + x_0) - f(y/t)$. Since y/t ranges over all of Y, we need

$$c \le \inf_{u \in Y} \big[p(u + x_0) - f(u)\big]. \tag{1}$$

For $t < 0$: setting $s = -t > 0$, we need $f(y) - sc \le p(y - sx_0)$, i.e., $c \ge f(y/s) - p(y/s - x_0)$. Since y/s ranges over Y:

$$c \ge \sup_{v \in Y} \big[f(v) - p(v - x_0)\big]. \tag{2}$$

We must verify that the lower bound does not exceed the upper bound. For all $u, v \in Y$:

$$f(u) + f(v) = f(u + v) \le p(u + v) = p((u + x_0) + (v - x_0)) \le p(u + x_0) + p(v - x_0),$$

so $f(v) - p(v - x_0) \le p(u + x_0) - f(u)$. Taking the supremum over v and infimum over u confirms compatibility. Choose c in the interval $[\sup_v(f(v) - p(v - x_0)),\ \inf_u(p(u + x_0) - f(u))]$.

Step 2: Zorn's Lemma. Let $\mathcal{F}$ be the set of all pairs (Z, g) where $Y \subset Z \subset X$ is a subspace and $g\colon Z \to \mathbb{R}$ is linear with $g|_Y = f$ and $g(z) \le p(z)$ for all $z \in Z$. Order $\mathcal{F}$ by extension: $(Z_1, g_1) \le (Z_2, g_2)$ if $Z_1 \subset Z_2$ and $g_2|_{Z_1} = g_1$.

Every chain in $\mathcal{F}$ has an upper bound: take the union of the subspaces and the common extension. By Zorn's Lemma, $\mathcal{F}$ has a maximal element (Z^*, F). If $Z^* \ne X$, Step 1 would produce a proper extension, contradicting maximality. Hence $Z^* = X$. □

Corollary 3.3 (Extension of Bounded Linear Functionals)**.** *Let Y be a subspace of a normed space X and $f \in Y^*$. Then there exists $F \in X^*$ with $F|_Y = f$ and $\|F\|_{X^*} = \|f\|_{Y^*}$.*

Proof. Apply Theorem 3.2 with $p(x) = \|f\| \cdot \|x\|$. Since $f(y) \le |f(y)| \le \|f\|\|y\| = p(y)$, the hypothesis is satisfied. The extension F satisfies $F(x) \le \|f\|\|x\|$ and $F(-x) \le \|f\|\|x\|$, giving $|F(x)| \le \|f\|\|x\|$, so $\|F\| \le \|f\|$. Since F extends f, $\|F\| \ge \|f\|$. □

Corollary 3.4 (Existence of Norming Functionals)**.** *For any $x_0 \ne 0$ in a normed space X, there exists $f \in X^*$ with $\|f\| = 1$ and $f(x_0) = \|x_0\|$.*

Proof. Define $g\colon \mathbb{R}x_0 \to \mathbb{R}$ by $g(\alpha x_0) = \alpha\|x_0\|$. Then g is linear with $\|g\| = 1$ and $g(x_0) = \|x_0\|$. Extend by Corollary 3.3. □

Hahn–Banach Theorem – Summary of Consequences

- Bounded linear functionals on subspaces extend to the whole space with preserved norm.
- For every nonzero x, there exists a norming functional $f \in X^*$ with $f(x) = \|x\|$ and $\|f\| = 1$.
- The dual X^* separates points: if $f(x) = f(y)$ for all $f \in X^*$, then $x = y$.
- Closed subspaces can be separated from external points by continuous linear functionals.

4 The Uniform Boundedness Principle

The Uniform Boundedness Principle converts pointwise boundedness of a family of operators into uniform boundedness. Its proof relies on the Baire Category Theorem (Lesson 18, Theorem 7.1).

Theorem 4.1 (Baire Category Theorem – Recall)**.** *Let (X, d) be a complete metric space. If $X = \bigcup_{n=1}^{\infty} F_n$ where each F_n is closed, then at least one F_n has nonempty interior. Equivalently, a countable intersection of open dense sets is dense.*

We do not reprove the Baire Category Theorem here; the complete proof is given in Lesson 18.

Theorem 4.2 (Uniform Boundedness Principle / Banach–Steinhaus)**.** *Let X be a Banach space, Y a normed space, and $\{T_\alpha\}_{\alpha \in A} \subset \mathcal{B}(X, Y)$ a family of bounded linear operators. If $\sup_\alpha \|T_\alpha x\| < \infty$ for every $x \in X$ (pointwise bounded), then $\sup_\alpha \|T_\alpha\| < \infty$ (uniformly bounded).*

Proof. For each $n \in \mathbb{N}$, define the closed set

$$F_n = \{x \in X : \|T_\alpha x\| \leq n \text{ for all } \alpha \in A\} = \bigcap_{\alpha \in A} \{x : \|T_\alpha x\| \leq n\}.$$

Each F_n is closed (as an intersection of closed sets, since each $x \mapsto \|T_\alpha x\|$ is continuous). Since the family is pointwise bounded, for each x there exists n with $x \in F_n$, so $X = \bigcup_{n=1}^{\infty} F_n$.

By the Baire Category Theorem, some F_N has nonempty interior: there exist $x_0 \in X$ and $r > 0$ with $B(x_0, r) \subset F_N$.

For any $x \in X$ with $\|x\| \leq 1$, we have $x_0 + rx \in B(x_0, r) \subset F_N$ and $x_0 \in F_N$, so for all α:

$$\|T_\alpha(rx)\| = \|T_\alpha(x_0 + rx) - T_\alpha(x_0)\| \leq \|T_\alpha(x_0 + rx)\| + \|T_\alpha(x_0)\| \leq N + N = 2N.$$

Thus $\|T_\alpha x\| \leq 2N/r$ for all $\|x\| \leq 1$, giving $\|T_\alpha\| \leq 2N/r$ for all α. □

Corollary 4.3 (Pointwise Convergence of Operators)**.** *Let X be a Banach space, Y a normed space, and $(T_n)_{n=1}^{\infty} \subset \mathcal{B}(X, Y)$ a sequence such that $T_n x \to Tx$ for every $x \in X$ (pointwise convergence to some T). Then:*

(i) $\sup_n \|T_n\| < \infty$.

(ii) T is a bounded linear operator.

(iii) $\|T\| \leq \liminf_{n \to \infty} \|T_n\|$.

Proof. **(i)** For each x, the convergent sequence $(T_n x)$ is bounded: $\sup_n \|T_n x\| < \infty$. By Theorem 4.2, $\sup_n \|T_n\| < \infty$.

(ii) Linearity of T follows from passing limits through addition and scalar multiplication. For boundedness: let $M = \sup_n \|T_n\|$. Then $\|T_n x\| \le M\|x\|$ for all n, and taking $n \to \infty$ gives $\|Tx\| \le M\|x\|$.

(iii) For any x with $\|x\| \le 1$: $\|Tx\| = \lim_n \|T_n x\| \le \liminf_n \|T_n\| \cdot \|x\| \le \liminf_n \|T_n\|$. Taking the supremum over x gives $\|T\| \le \liminf_n \|T_n\|$. □

Uniform Boundedness in Value Function Approximation

In temporal-difference learning, one often considers a sequence of approximate Bellman operators T_n (e.g., using n samples or truncated to n steps). If $T_n V \to TV$ pointwise for each value function V, the Uniform Boundedness Principle guarantees that $\sup_n \|T_n\| < \infty$. This uniform bound is essential for proving that the approximation error does not blow up: if $\|T_n V - T_n W\| \le M\|V - W\|$ uniformly in n, then the approximate operators are equi-Lipschitz and one can safely pass limits through compositions.

5 The Open Mapping Theorem

Theorem 5.1 (Open Mapping Theorem)**.** *Let X and Y be Banach spaces and $T\colon X \to Y$ a bounded linear operator that is surjective. Then T is an open map: $T(U)$ is open in Y whenever U is open in X.*

Proof. Write $B_X = B_X(0,1)$ (open unit ball in X) and $B_Y = B_Y(0,1)$.

Step 1: $\overline{T(B_X)}$ contains a ball about 0. Since T is surjective, $Y = \bigcup_{n=1}^{\infty} T(nB_X) = \bigcup_{n=1}^{\infty} n \cdot T(B_X)$. By the Baire Category Theorem (Theorem 4.1), some $\overline{n \cdot T(B_X)}$ has nonempty interior. By scaling, $\overline{T(B_X)}$ has nonempty interior: there exist $y_0 \in Y$ and $s > 0$ with $B_Y(y_0, s) \subset \overline{T(B_X)}$.

We claim $B_Y(0,s) \subset \overline{T(B_X)}$. Since $T(B_X)$ is symmetric ($y \in T(B_X)$ implies $-y \in T(B_X)$, as $Tx \in T(B_X)$ gives $T(-x) = -Tx \in T(B_X)$), we have $B_Y(-y_0, s) \subset \overline{T(B_X)}$ as well.

For any y with $\|y\| < s$, both $y + y_0$ and $-y + y_0$ lie in $\overline{T(B_X)}$. Since $T(B_X) + T(B_X) \subset T(2B_X)$, we get $\overline{T(B_X)} + \overline{T(B_X)} \subset \overline{T(2B_X)}$. In particular, $(y + y_0) + (-y + y_0) - 2y_0$... let us argue more directly. We have $(y + y_0) - (-y + y_0) = 2y$, and since both points are in $\overline{T(B_X)}$, we get $2y \in \overline{T(B_X)} - \overline{T(B_X)} \subset \overline{T(2B_X)}$, hence $y \in \overline{T(B_X)}$. So $B_Y(0,s) \subset \overline{T(B_X)}$.

Step 2: Opening the closure. We show $\overline{T(B_X)} \subset T(2B_X)$. Take $y \in \overline{T(B_X)}$. From Step 1, for each $r > 0$, $B_Y(0, rs) \subset \overline{T(rB_X)}$.

Choose $x_1 \in B_X$ with $\|y - Tx_1\| < s/2$. Then $y - Tx_1 \in B_Y(0, s/2) \subset \overline{T(\frac{1}{2}B_X)}$, so choose $x_2 \in \frac{1}{2}B_X$ with $\|y - Tx_1 - Tx_2\| < s/4$. Inductively, choose $x_k \in 2^{-(k-1)}B_X$ with $\|y - \sum_{j=1}^{k} Tx_j\| < s \cdot 2^{-k}$.

Then $\sum_k \|x_k\| < 1 + 1/2 + 1/4 + \cdots = 2$. Since X is complete, $x = \sum_{k=1}^{\infty} x_k$ converges with $\|x\| < 2$, and $Tx = \lim_{k\to\infty} T\big(\sum_{j=1}^{k} x_j\big) = y$. So $y \in T(2B_X)$.

Step 3: Combining. $B_Y(0,s) \subset \overline{T(B_X)} \subset T(2B_X)$, so $B_Y(0, s/2) \subset T(B_X)$.

Step 4: T is open. Let $U \subset X$ be open and $y_0 = Tx_0 \in T(U)$. Choose $r > 0$ with $B_X(x_0, r) \subset U$. Then $T(U) \supset T(B_X(x_0, r)) = Tx_0 + T(rB_X) \supset y_0 + B_Y(0, rs/2)$, so $T(U)$ is open. □

Corollary 5.2 (Bounded Inverse Theorem)**.** *If $T\colon X \to Y$ is a bounded linear bijection between Banach spaces, then $T^{-1}\colon Y \to X$ is bounded.*

Proof. T is an open map by Theorem 5.1, so T^{-1} is continuous: preimages of open sets under T^{-1} are images of open sets under T, hence open. □

Open Mapping Theorem – Key Consequences

- A bounded linear bijection between Banach spaces has a bounded inverse (so "algebraic isomorphism + continuity" implies "topological isomorphism").
- If a vector space is complete under two norms and one dominates the other, the norms are equivalent.
- A surjective bounded operator T sends open sets to open sets, and in particular $T(B_X)$ contains a ball about 0.

6 The Closed Graph Theorem

Definition 6.1 (Closed Graph)**.** A linear operator $T\colon X \to Y$ between normed spaces has a *closed graph* if its graph $G(T) = \{(x, Tx) : x \in X\} \subset X \times Y$ is closed in the product topology (equivalently, in the norm $\|(x, y)\| = \|x\| + \|y\|$). The closed graph condition is:

$$\text{if } x_n \to x \text{ in } X \text{ and } Tx_n \to y \text{ in } Y, \text{ then } y = Tx.$$

Remark 6.2. Every bounded (continuous) operator has a closed graph: if $x_n \to x$ and $Tx_n \to y$, then $Tx_n \to Tx$ by continuity, so $y = Tx$ by uniqueness of limits. The Closed Graph Theorem is the converse for Banach spaces.

Theorem 6.3 (Closed Graph Theorem)**.** *Let X and Y be Banach spaces. A linear operator $T\colon X \to Y$ is bounded if and only if it has a closed graph.*

Proof. ($\Rightarrow$) Already noted in Remark 6.2.

($\Leftarrow$) Assume $G(T)$ is closed. The product space $X \times Y$ with the norm $\|(x, y)\| = \|x\| + \|y\|$ is a Banach space (since X and Y are both complete). Since $G(T)$ is a closed subspace of a Banach space, $G(T)$ is itself a Banach space.

Define the coordinate projections:

$$\pi_1\colon G(T) \to X, \quad \pi_1(x, Tx) = x,$$
$$\pi_2\colon G(T) \to Y, \quad \pi_2(x, Tx) = Tx.$$

Both π_1 and π_2 are bounded (by $\|x\| \le \|(x, Tx)\|$ and $\|Tx\| \le \|(x, Tx)\|$ respectively). The map π_1 is linear and bijective (since T is a function: each x determines a unique (x, Tx)).

By the Bounded Inverse Theorem (Corollary 5.2), $\pi_1^{-1}\colon X \to G(T)$ is bounded. Since $T = \pi_2 \circ \pi_1^{-1}$ and both π_2 and π_1^{-1} are bounded, T is bounded. $\square$

Remark 6.4 (Practical Usage). The Closed Graph Theorem is often the most convenient way to prove that an operator is bounded: instead of estimating $\|Tx\|$ directly, one only needs to verify the sequential closedness condition (if $x_n \to x$ and $Tx_n \to y$, then $y = Tx$), which is typically easier because one has two convergence hypotheses to work with instead of one.

Example 6.5 (Symmetric Operators are Bounded)**.** Let H be a Hilbert space and $T\colon H \to H$ a linear operator satisfying $\langle Tx, y\rangle = \langle x, Ty\rangle$ for all $x, y \in H$ (a symmetric operator defined on all of H). We show T is bounded via the Closed Graph Theorem.

Suppose $x_n \to x$ and $Tx_n \to y$. For any $z \in H$:

$$\langle y, z\rangle = \lim_{n\to\infty} \langle Tx_n, z\rangle = \lim_{n\to\infty} \langle x_n, Tz\rangle = \langle x, Tz\rangle = \langle Tx, z\rangle.$$

Since this holds for all z, we conclude $y = Tx$ (by the non-degeneracy of the inner product). The graph of T is closed, so T is bounded by the Closed Graph Theorem.

Closed Graph Theorem and Solution Map Continuity

Consider the parametric Bellman equation $V_\theta = T_\theta V_\theta$ where θ parameterizes the reward or transition kernel. The solution map $\theta \mapsto V_\theta$ is often linear (when the Bellman equation is linear in V, as in policy evaluation). To prove that this map is continuous—that small changes in θ lead to small changes in V_θ—one can use the Closed Graph Theorem. If $\theta_n \to \theta$ and $V_{\theta_n} \to W$, then by continuity of the Bellman operator in θ: $W = \lim T_{\theta_n} V_{\theta_n} = T_\theta W$, so $W = V_\theta$ by uniqueness of the fixed point. The graph is closed, hence the map is continuous.

7 Exercises

Exercise 7.1 (Hahn–Banach on Finite-Dimensional Spaces (*))**.** Let $X = \mathbb{R}^3$ and $Y = \{(x_1, x_2, 0) : x_1, x_2 \in \mathbb{R}\}$. Define $f\colon Y \to \mathbb{R}$ by $f(x_1, x_2, 0) = x_1 + 2x_2$. Using the Hahn–Banach theorem (with $p(x) = \|f\|_{Y^*} \cdot \|x\|_2$), find all extensions $F \in (\mathbb{R}^3)^*$ with $\|F\| = \|f\|$.
Hint: First compute $\|f\|_{Y^*}$, then write $F(x) = ax_1 + bx_2 + cx_3$ and determine a, b, c.

Exercise 7.2 (Separation of a Point from a Closed Subspace (*))**.** Let $M = \{(x_n) \in \ell^2 : x_1 = 0\}$ and $x_0 = (1, 0, 0, \ldots) \in \ell^2$. Explicitly construct a functional $f \in (\ell^2)^*$ with $f|_M = 0$ and $f(x_0) = \operatorname{dist}(x_0, M)$. Verify $\|f\| = 1$.

Exercise 7.3 (Norming Functional in L^p (**))**.** For $f \in L^p([0, 1])$ with $f \neq 0$ and $1 < p < \infty$, explicitly construct the norming functional $g \in (L^p)^* = L^q$ (where $1/p + 1/q = 1$) satisfying $\|g\|_q = 1$ and $\int_0^1 fg\,dx = \|f\|_p$.
Hint: Try $g = c|f|^{p-2}f$ for an appropriate constant c.

Exercise 7.4 (Uniform Boundedness for Fourier Coefficients (**))**.** Let $\{e_n\}_{n=1}^\infty$ be an orthonormal sequence in a Hilbert space H. Define $T_n\colon H \to \mathbb{R}$ by $T_n(x) = \langle x, e_n\rangle$.

(a) Show that each T_n is bounded with $\|T_n\| = 1$.

(b) Show that the family $\{T_n\}$ is pointwise bounded (use Bessel's inequality from Lesson 20, Theorem **??**).

(c) Apply the Uniform Boundedness Principle to conclude $\sup_n \|T_n\| < \infty$. Why is this trivial in this case?

(d) Give an example where the UBP gives a non-trivial conclusion that is not obvious from direct computation.

Exercise 7.5 (Open Mapping Failure Without Completeness (**))**.** Let $X = (C^1[0, 1], \|\cdot\|_\infty)$ and $Y = (C[0, 1], \|\cdot\|_\infty)$. Define $T\colon X \to Y$ by $Tf = f'$ (differentiation).

(a) Show that T is linear, surjective (onto Y), but *not* bounded with respect to the given norms.

(b) Explain why this does not contradict the Open Mapping Theorem.

(c) What changes if we equip X with the norm $\|f\|_{C^1} = \|f\|_\infty + \|f'\|_\infty$?

Exercise 7.6 (Closed Graph Theorem Application (**))**.** Let X be a Banach space and $T\colon X \to X$ a linear operator such that $\langle f, Tx\rangle = \langle g, x\rangle$ for all $x \in X$, where $f, g \in X^*$ are fixed nonzero functionals and $\langle\cdot,\cdot\rangle$ denotes the duality pairing. Show that T is bounded by verifying the closed graph condition.

Exercise 7.7 (Banach–Steinhaus and Divergence of Fourier Series (***)). Let $S_n \colon C[-\pi, \pi] \to \mathbb{R}$ be the n-th partial sum of the Fourier series evaluated at 0:

$$S_n(f) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t)\, D_n(t)\, dt,$$

where $D_n(t) = \sum_{k=-n}^{n} e^{ikt}$ is the Dirichlet kernel.

(a) Show $\|S_n\| = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_n(t)|\, dt$ (use the characterization of the norm of a functional defined by integration against a kernel).

(b) Show $\|S_n\| \to \infty$ as $n \to \infty$ (it is known that $\|S_n\| \sim \frac{4}{\pi^2} \log n$).

(c) Apply the Uniform Boundedness Principle to conclude: there exists $f \in C[-\pi, \pi]$ such that $\sup_n |S_n(f)| = \infty$, i.e., the Fourier series of f diverges at 0.

Exercise 7.8 (Dual of ℓ^1 via Hahn–Banach (***)). We know $(\ell^1)^* = \ell^\infty$ (isometrically). Use the Hahn–Banach theorem to prove one direction: every $a = (a_n) \in \ell^\infty$ defines an element of $(\ell^1)^*$ with norm $\|a\|_\infty$, and conversely, given $f \in (\ell^1)^*$, the sequence $a_n = f(e_n)$ lies in ℓ^∞ with $\|a\|_\infty = \|f\|$, where e_n denotes the n-th standard unit vector.

8 Summary

- **Hahn–Banach Theorem:** A linear functional dominated by a sublinear functional on a subspace extends to the whole space with the domination preserved. For normed spaces: bounded linear functionals extend with preserved norm. Consequences include existence of norming functionals and separation of points from closed subspaces.
- **Uniform Boundedness Principle (Banach–Steinhaus):** A pointwise bounded family of bounded linear operators on a Banach space is uniformly bounded. Requires completeness of the domain (via Baire). Consequence: pointwise convergent operator sequences have bounded limits.
- **Open Mapping Theorem:** A surjective bounded linear operator between Banach spaces is an open map. Consequence: a bounded linear bijection has a bounded inverse (Bounded Inverse Theorem), and completeness under a dominating norm implies norm equivalence.
- **Closed Graph Theorem:** A linear operator between Banach spaces is bounded if and only if its graph is closed. This provides a convenient sufficient condition for boundedness: one only needs to check the sequential closedness condition.
- **Common hypotheses:** The Hahn–Banach theorem requires no completeness. The remaining three theorems require the domain (and for the Open Mapping Theorem, also the codomain) to be complete, via the Baire Category Theorem.
- **RL connections:** Hahn–Banach guarantees dual variables in constrained MDPs; the UBP ensures uniform bounds on operator families arising in iterative algorithms; the Closed Graph Theorem provides stability of solution maps under parameter perturbations.

9 Further Reading

For a detailed treatment of the four fundamental theorems with applications, see Brézis [2011], Chapters 1–2, and Rudin [1991], Chapters 2–4. The Hahn–Banach theorem in the complex case (which requires a slightly different proof) is given in Kreyszig [1989], Section 4.3. For applications to optimization and duality in the MDP setting, see Puterman [2014], Chapter 6.

The preceding lesson (Lesson 20) developed projections and the Riesz Representation Theorem in Hilbert spaces. The next lesson (Lesson 22) develops fixed point theorems—Banach, Brouwer, Schauder, and Kakutani—which provide existence and convergence guarantees for the iterative algorithms of reinforcement learning.

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